06.10.25

Data

observed | synthetic

observed on field; recorded using instrument

- data which we create using source (impulse) / source signature
- based on medium properties
- synthetic data is matched with observed data - doesn't match? recompute again and again

Delineate Earth's structure —

Receiver Functions (RF)

↳ data has 3 components (two horizontal, 1 vertical)

rotate data (twice) : R & T components → L and Q components — backazimuth recorded data to be downloaded from IRIS/Wilber

We deal with teleseismic data - deconvolution operation

cd RF

↓

ls → cd ex-prf

↓

conda activate seispy

↓

cat Readme

gedit/cat rf.cfg (details of rf file; station and configurations, event specifications)

prf rf.cfg (downloads eq. data online from specifications)

pickrf RFresult/CB.NJ2 (shows R and T components, distance and backazimuth)

plotrf RFresult/CB.NJ2 (saves as png file)
↳ Chinese station

peak @ 0 → first P arrival → next '+ve' peak → converted phase : $\Delta t \times V_p$ (~ 8km/s) = moho dist.
→ removed in LQT

To remove unwanted signal → select → next → finish

Then plotrf

peak shifted from zero → thin sedimentary layer

13.10.25

Pyraysum

Open terminal → conda activate pyray
↓
cd /home/GG/PyRaysum/pyraysum/examples/notebooks

↓

open jupyter notebook

↓

(2 reproduce Porter - 2011)

open an example and go to 'Kernel' tab

↓

click 'Change Kernel' option and select 'Python (pyraysum)'

↓

run each cell - change parameters.
(shift + enter)

Code Details (run each cell):-

- import parameters prs, Geometry, Model, Control
- 2 layer half-space model (2 layers + half space)
; define other parameters

- rotation type :- 'PVH' = $P S_V S_H$; use rot = 1 for RTZ
may use 2 multiples ; try for RTZ and PVH.

PVH → no energy @ 0 PVH → 2 mul

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anisotropy → 180° periodicity (2 lobes) dip → 360° periodicity (1 lobe)
↳ we have to check 360° ...

(-) → slow axis

change parameters in anisotropy section
(don't put everything in model.change)

27.10.25

- The last location of earthquake obtained found out is stored in 'hyp.out'

- copy all miniseed files or merged files in a new folder.

↓

split if you have taken the merged file

↓

convert to 'sac' format : mseed2sac *BH*
(*BH*: common part in all file names)

→ to split : seisei → '2' to split → file name

↓

remove miniseed files once SAC files are created
— rm * 001-BH*

↓

type 'sac'

↓

r <file name> → p : to plot → 'q' to quit

↓

To view header file type 'lh' [least header]

Components of data file/header: B – starting time of waveform in seconds
if origin is shifted to right, B becomes -ve.

E - end time (in s) of waveform

DELTA - sampling rate (higher frequency requires higher sampling rate)

In 1 s here, we have 50 data points/samples to reconstruct our original waveform from digital.

KZTIME - starting time of waveform
it is the time of '0' on waveform (might not be B/at the origin)

KSTNM - station name

KCMPNM - components

KNETWK - network

We have to update the event lat. and long and station lat-long and elevation:

ch EVLA < > EVLO <> EVDP <>
(change header)

header file variable list provided in 'sac-manual.pdf'

update station latitude and longitude

ch STLA <> STLO <> STEL <>

wh (write header to save information)

SAC - Seismic Analysis Code

Apply bandpass filter:

bp p 2 n 4 c 1 5

poles ~~poles~~ ~~poles~~ n. of nodes ~~poles~~ corners Lf Hf

'o' to zoom out

rtr (remove ~~train~~ – long term signals removed)

To pick 'p' arrivals type 'ppk' then 'p' to mark; 's' to mark S-arrival, 'q' to quit

wh → lh

28/10/25

- Find the origin time of the event from 'hyp.out' (it is contained in heading)
- Find its difference with KZTIME (time of 'zero' on graph)
- Mark new origin.

CHNHDR O GMT $year $julday $hour $min $sec O

Task – compare picked P & S arrival time and TauP arrival times saved using user variables